

MMP 1
Serie 1
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1 Aufgabe 1

1.a

$$\begin{aligned}
 (\widehat{Tf})_k &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} T(e^{-\frac{2\pi i}{N}jk} f_j) \\
 &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{\frac{2\pi i}{N}jk} f_{-j} \\
 &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-\frac{2\pi i}{N}(-j)k} f_{-j} \\
 &= \hat{f}_{-k} = (T\hat{f})_k
 \end{aligned} \tag{1.1}$$

1.b

$$\begin{aligned}
 \overline{\hat{f}}_k &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \overline{e^{-\frac{2\pi i}{N}jk} f_j} \\
 &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{\frac{2\pi i}{N}jk} \overline{f_j} \\
 &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-\frac{2\pi i}{N}(-j)k} \overline{f_j} \\
 &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-\frac{2\pi i}{N}j(-k)} \overline{f_j} \\
 &= \hat{\hat{f}}_{-k} = T\hat{\hat{f}} \\
 &\stackrel{(1.1)}{=} \widehat{T\hat{f}}
 \end{aligned} \tag{1.2}$$

1.c

$$\begin{aligned}
 (\widehat{Sf})_k &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} S(e^{-\frac{2\pi i}{N}jk} f_j) \\
 &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-\frac{2\pi i}{N}(j+1)k} f_{j+1} \\
 &= e^{\frac{2\pi i}{N}k} \frac{1}{\sqrt{N}} \sum_{l=0}^{N-1} e^{-\frac{2\pi i}{N}lk} f_l \\
 &= e^{\frac{2\pi i}{N}k} \hat{f}_k
 \end{aligned} \tag{1.3}$$

2 Aufgabe 2

2.a

$S \in M_{n \times n}(\mathbb{C})$ hat die Form:

$$(S)_{ij} = (S)_{i+1,j+1} \quad \text{mit} \quad (S)_{1,j} = \delta(1,j) \tag{2.1}$$

dabei wird mit mod_n gerechnet.

$$\begin{aligned}
 (AS)_{i,j} &= \sum_{k=0}^{n-1} a_{ik} s_{kj} \\
 &= \sum_{k=0}^{n-1} a_{ik} \delta(j+1, k) \\
 &= (A)_{i,j+1}
 \end{aligned} \tag{2.2}$$

$$\begin{aligned}
(SA)_{ij} &= \sum_{k=0}^{n-1} s_{ik} a_{kj} \\
&= \sum_{k=0}^{n-1} \delta(i, k-1) a_{kj} \\
&= (A)_{i-1, j}
\end{aligned} \tag{2.3}$$

$$\Rightarrow (A)_{i-1, j} = (A)_{i, j+1} \Leftrightarrow (A)_{ij} = (A)_{i+1, j+1} \tag{2.4}$$

2.b

Let $a \in \mathbb{C}^n$ be the first row of A .

$$\begin{aligned}
(Af)_k &= \sum_{j=0}^{n-1} a'_{kj} f_j \\
&= \sum_{j=0}^{n-1} a'_{k-1, j-1} f_j \\
&= \sum_{j=0}^{n-1} a'_{k-k, j-k} f_j \\
&= \sum_{j=0}^{n-1} a_{j-k} f_j
\end{aligned} \tag{2.5}$$

Define $b \in \mathbb{C}^n$ such that $a_{j-k} = b_{k-j}$

$$\Rightarrow (Af)_k = \sum_{j=0}^{n-1} b_{k-j} f_j = (b \star f)_k \tag{2.6}$$

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3.a

$$\begin{aligned}
\lim_{n \rightarrow \infty} f_{n(x)}^1 &= 0 \\
\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n^1(x) dx &= \lim_{n \rightarrow \infty} \int_n^{n+1} dx = 1
\end{aligned} \tag{3.1}$$

$$\begin{aligned}
\lim_{n \rightarrow \infty} f_{n(x)}^2 &= \lim_{n \rightarrow \infty} \begin{cases} \frac{1}{n} & x \in [0, n] \\ 0 & x \notin [0, n] \end{cases} = 0 \\
\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n^2 dx &= \lim_{n \rightarrow \infty} \frac{1}{n} \int_0^n dx = \lim_{n \rightarrow \infty} \frac{n}{n} = 1
\end{aligned} \tag{3.2}$$

$$\begin{aligned}
\lim_{n \rightarrow \infty} f_{n(x)}^3 &= \lim_{n \rightarrow \infty} \begin{cases} n & x \in [\frac{1}{n}, \frac{2}{n}] \\ 0 & x \notin [\frac{1}{n}, \frac{2}{n}] \end{cases} = 0 \\
\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n^3 dx &= \lim_{n \rightarrow \infty} n \int_{\frac{1}{n}}^{\frac{2}{n}} dx = \lim_{n \rightarrow \infty} n \frac{2-1}{n} = 1
\end{aligned} \tag{3.3}$$

$$\begin{aligned}
\lim_{n \rightarrow \infty} f_{n(x)}^4 &= \lim_{n \rightarrow \infty} \begin{cases} n & x \in [0, \frac{1}{n}] \\ 0 & x \notin [0, \frac{1}{n}] \end{cases} = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases} \\
\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f_n^4 dx &= \lim_{n \rightarrow \infty} n \int_0^{\frac{1}{n}} dx = \frac{n}{n} = 1
\end{aligned} \tag{3.4}$$

3.b

$$\begin{aligned}
f(x)_n &= \begin{cases} ne^{-\frac{n(1+x)}{x}} & x \neq 0 \\ 0 & x = 0 \end{cases} \\
\int_0^1 f_n(x) dx &= \left[\frac{n}{n} x^2 e^{-\frac{n}{x} + n} \right]_0^1 = 1, \quad \forall n \in \mathbb{N}
\end{aligned} \tag{3.5}$$

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4.a

$$\begin{aligned}\lim_{n \rightarrow \infty} \int_0^\pi (1 - \sin(x))^n dx &= \int_0^\pi \lim_{n \rightarrow \infty} (1 - \sin(x))^n dx \\ &= \int_0^\pi \chi_{\{0, \pi\}}(x) dx = 0\end{aligned}\tag{4.1}$$

4.b

$$\begin{aligned}\lim_{n \rightarrow \infty} \int_{-\infty}^\infty \frac{n \sin(\frac{x}{n})}{x^3 + x} dx \\ &= \int_{-\infty}^\infty \lim_{n \rightarrow \infty} \frac{n \left(\frac{x}{n} - \frac{x^3}{6n^3} + O\left(\left(\frac{x}{n}\right)^5\right) \right)}{x^2 + x} dx \\ &= \int_{-\infty}^\infty \frac{1}{x^2 + 1} dx = [\tan^{-1}(x)]_{-\infty}^\infty = \frac{\pi}{2} + \frac{\pi}{2} = \pi\end{aligned}\tag{4.2}$$

4.c

$$\lim_{n \rightarrow \infty} \int_{-\infty}^\infty n \log\left(1 + \frac{f(x)}{n}\right) dx = \int_{-\infty}^\infty \lim_{n \rightarrow \infty} n \left(\frac{f(x)}{n} + O\left(\left(\frac{f(x)}{n}\right)^2\right) \right) dx = \int_{-\infty}^\infty f(x) dx\tag{4.3}$$

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5.a

$$\begin{aligned}\lim_{n \rightarrow \infty} \int_1^2 \frac{n}{1 + nx^2} dx &= \int_1^2 \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{n} + x^2} dx \\ &= \left[-\frac{1}{x} \right]_1^2 = \frac{1}{2}\end{aligned}\tag{5.1}$$

5.b

$$\lim_{x \rightarrow 0} \int_0^\infty \frac{e^{-xt}}{1 + t^2} dt = \int_0^\infty \frac{1}{1 + t^2} dt = \frac{\pi}{2}\tag{5.2}$$